EXPLORATORY DATA ANALYSIS REPORT

ICU Mechanical Ventilation Risk Prediction

eICU Collaborative Research Database Demo v2.0.1

Objective: Build a binary classification model to predict whether an ICU patient will require mechanical ventilation in the next 6 to 12 hours, using time-series clinical vitals, respiratory charting, and treatment data.

Component	Detail
Target Variable	y_high = 1 if hour falls in the 6-12h window before ventilation start
Dataset	eICU Collaborative Research Database Demo v2.0.1
Patients	2,520 ICU stays across 5 tables
Final Dataset	136,867 patient-hours with 28 features
Positive Rate	0.22% (298 danger hours out of 136,867)
Notebook	1_0_bg_EDA.ipynb

Contents

1. Dataset Overview

The eICU Collaborative Research Database Demo is a multi-centre critical care database containing de-identified patient data from intensive care units across the United States. All timestamps are recorded as offsets in minutes from each patient's ICU admission, making it a time-series dataset.

Five tables were used in this project:

Table	Rows	Columns	Role in Project
patient.csv	2,520	29	Patient demographics and stay metadata
respiratoryCare.csv	5,436	34	Ventilation events — used to build labels ONLY
respiratoryCharting.csv	176,089	7	Long-format respiratory vitals (FiO2, PEEP, RR, etc.)
vitalPeriodic.csv	1,634,960	19	Periodic vital signs (HR, SpO2, respiration rate)
treatment.csv	38,290	5	Clinical treatments applied during stay

WHY

vitalAperiodic.csv was excluded. Although it contains non-invasive blood pressure readings (~88% present), vitalPeriodic already provides the three most clinically relevant signals for ventilation prediction. Adding more tables at this stage would increase complexity without a proven benefit. It can be added in a future iteration if model performance is insufficient.

1.1 Time Encoding

All time columns across all tables were renamed to a unified column t_min representing minutes elapsed since ICU admission. This simplifies alignment across tables and prevents confusion when merging. Negative offsets (readings before admission, likely data entry errors) were dropped.

1.2 Key Observations from Data Overview

- **age** is stored as a string (object dtype) in the patient table because of '> 89' values representing patients over 89. This requires special handling before numeric operations.
- **admissionheight** has a maximum value of 600 cm — physiologically impossible, indicating data entry errors.
- **admissionweight** has a maximum value of 515 kg — also impossible.
- **dischargeweight** is 51% missing and is a post-ICU measurement — not predictive for ventilation. Dropped entirely.
- **ventendoffset** in respiratoryCare is entirely zero across all rows. Completely useless, dropped.
- **Five columns** in respiratoryCare are 100% empty: sighpreslimit, lowironoxlimit, highironoxlimit, meanairwaypreslimit, setapneaie. Dropped.
- **vitalPeriodic** contains 16 columns but only heartrate, sao2, and respiration are sufficiently complete (85-99% present). All others are 86-99% missing.

2. Missing Value Analysis

Missing value analysis was conducted for each table, computing the percentage of missing values per column. Results were visualized using missingno matrix plots to identify whether missingness was random (white bars distributed evenly) or systematic (whole columns empty).

Column	Table	Missing %	Decision & Reason
dischargeweight	patient	51%	Dropped — post-ICU, not available at prediction time
hospitaladmitsource	patient	23.6%	Imputed with mode — categorical, moderate missing
apacheadmissiondx	patient	11.9%	Imputed with mode — categorical, moderate missing
admissionweight	patient	7.9%	Imputed with median — numeric, low missing
admissionheight	patient	2.7%	Imputed with median after outlier removal
sao2	vitalPeriodic	11.8%	Kept — aligned via merge_asof, missingness is informative
respiration	vitalPeriodic	15.5%	Kept — aligned via merge_asof, missingness is informative
temperature	vitalPeriodic	93.1%	Dropped — too sparse, requires invasive monitoring
cvp	vitalPeriodic	87.6%	Dropped — too sparse, central line required
etco2, st1/2/3, icp	vitalPeriodic	95-99%	Dropped — near-completely missing

The high missingness in most vitalPeriodic columns reflects clinical reality: measurements like central venous pressure (CVP) and intracranial pressure (ICP) require invasive monitoring lines that are only placed for specific patient subsets.

KEY INSIGHT

Missing values in the final engineered features (fio2_last, heartrate_last, sao2_last) were NOT imputed before modeling. Instead, binary missingness indicator columns were created (fio2_missing, hr_missing, sao2_missing). Infrequent measurement may reflect lower patient acuity or monitoring instability — both clinically informative signals.

3. Outlier Treatment

Physiologically impossible values were identified during univariate analysis. These are real data quality issues — not statistical outliers — and must be removed rather than capped or transformed. The strategy was to replace impossible values with NaN and then impute with the column median.

Feature	Impossible Values Removed	Clinical Rationale
admissionheight	< 100 cm or > 250 cm	Adult human height range. Max 600 cm was clearly a data entry error.
admissionweight	< 20 kg or > 300 kg	Values above 300 kg are implausible without documentation.
heartrate	< 20 or > 250 bpm	Outside the compatible-with-life heart rate range.
sao2	< 50% or > 100%	SpO2 below 50% is incompatible with life; above 100% is impossible.
respiration	< 4 or > 60 breaths/min	Normal adult range; values of 0 and 188 indicate sensor errors.

Patient table outliers were treated immediately after loading. Vital sign outliers were treated during the merge_asof alignment step in feature engineering, ensuring impossible readings never enter the hourly grid.

4. Duplicate Handling

All five tables were checked for exact duplicate rows (all column values identical).

Table	Duplicates Found	Action
patient	0	No action needed
respiratoryCare	Not checked for removal	Time-series rows: multiple readings valid
respiratoryCharting	Not checked for removal	Long-format: same label at same time is valid
vitalPeriodic	Not checked for removal	Periodic readings: duplicates may be real
treatment	Not checked for removal	Event log: same treatment at same time is valid

The patient table was the only one where exact duplicates are definitionally impossible in valid data — each row should represent a unique ICU stay. Time-series and event tables intentionally retain all rows, as repeated entries may represent valid clinical documentation by multiple staff members.

5. Target Variable Analysis

Understanding when patients got ventilated — and how rare that event was — was essential for defining the prediction problem and anticipating modeling challenges.

Metric	Value	Implication
Total ICU patients	2,520	Dataset size
Ventilated patients	69 (2.7%)	Severe class imbalance at patient level
Median time to ventilation	1,619 min (~27 hours)	Half of patients ventilated in first day
Minimum time to ventilation	11 minutes	Some patients ventilated on admission
Maximum time to ventilation	29,563 min (~20 days)	Long tail of late ventilations
y_high = 1 rows (hourly grid)	298 of 136,867 (0.22%)	Extreme imbalance at row level

The 0.22% positive rate at the hourly level is more extreme than the 2.7% patient-level rate because most ventilated patients have only 6 labeled danger hours (the 6-12h window produces 6 hourly checkpoints), while their full ICU stay spans hundreds of safe hours.

IMPLICATION

Standard accuracy is meaningless here. A model predicting y=0 for every row achieves 99.78% accuracy while catching zero patients. AUPRC (Area Under the Precision-Recall Curve) must be used as the primary evaluation metric.

6. Univariate Analysis

Univariate distributions were examined for all candidate features. The goal was to understand each variable in isolation, identify unusual distributions, and decide whether to include it in the model.

6.1 Patient Demographics

- **Age:** Well-distributed, peaking around 65-70 years (typical ICU population). Retained as demographic baseline.
- **Gender:** More males (1,508) than females (1,008). Included as a categorical feature.
- **ICU Unit Type:** Dominated by Med-Surg ICU (1,898 patients). CSICU showed the highest ventilation rate (4.6%) despite smaller volume.

- **Admission Weight:** Right-skewed with outliers up to 515 kg. Retained after cleaning.
- **Admission Height:** Large spike at ~170 cm (normal) but values up to 600 cm. Retained after cleaning.

6.2 Vital Signs (vitalPeriodic)

- **Heart Rate:** Centered around 80-90 bpm with a right tail. 335 values at or below zero were removed as physiologically impossible.
- **SpO2 (sao2):** Strongly concentrated at 95-100%. 1,985 values below 50% removed. The concentration near 100% is expected — patients in ICU are actively managed.
- **Respiration Rate:** Centered around 15-25 breaths per minute. 11,472 values at zero and 1,515 values above 60 removed.

6.3 Respiratory Charting

The respiratoryCharting table is in long format — one row per measurement label and value. The top 20 labels by record count were examined:

Measurement Label	Record Count	Ventilated Patient Coverage	Decision
FiO2	27,432	96%	Include — strongest predictor
RR (patient)	22,210	35%	Skip — already in vitalPeriodic
PEEP	13,630	86%	Include — key pre-vent signal
Tidal Volume (set)	11,141	84%	Include
LPM O2	11,061	36%	Skip — low ventilated patient coverage
Vent Rate	10,576	—	Skip — ventilator setting, post-event
TV/kg IBW	10,711	—	Skip — computed from other values
SaO2 (from chart)	3,672	—	Skip — duplicate of vitalPeriodic sao2

7. Bivariate Analysis

Bivariate analysis compared feature distributions between ventilated and non-ventilated patients. This directly answered the question: does this variable look meaningfully different in patients who later needed ventilation?

7.1 Patient Demographics vs Ventilation Status

- **Age:** Nearly identical distribution between ventilated and non-ventilated patients. Weak linear predictor, but retained for clinical context and potential interaction effects.
- **Admission Weight:** No meaningful difference between groups. Retained as a contextual feature.
- **ICU Stay Length:** Ventilated patients stayed significantly longer (median ~200h vs ~50h). However, stay length is partially a consequence of ventilation, not a cause — it cannot be used as a feature.

7.2 FiO2 Before vs After Ventilation

A critical leakage check was performed: comparing FiO2 readings before vs after ventilation start for ventilated patients.

Signal	% Before Ventilation	% After Ventilation	Interpretation
FiO2	27%	73%	Mostly a ventilator setting post-event, but 27% are pre-event warnings
PEEP	27%	73%	Same pattern — majority post-event but meaningful pre-event signal
Tidal Volume (set)	30%	70%	Ventilator parameter — 30% pre-event readings

The pre-event readings are exactly what we need: a patient on supplemental oxygen or CPAP will have FiO2 and PEEP recorded before any intubation occurs. The merge_asof approach ensures that only pre-event readings are used at each patient-hour. This check confirmed that including FiO2 and PEEP is safe and meaningful.

7.3 ICU Unit Type Ventilation Rate

Ventilation rates varied across ICU types. CSICU (Cardiac Surgery ICU) showed the highest rate at 4.6%, while MICU (Medical ICU) showed a surprisingly low 1.4%. This variation justifies including unit type as a categorical feature.

7.4 Vital Signs vs Ventilation Status (Raw Data Limitation)

Vital sign distributions appeared similar between ventilated and non-ventilated patients when compared on the raw data. This is expected: the comparison includes all hours of a patient's stay, including safe hours long before any event. The meaningful signal only becomes visible when comparing danger hours (6-12h before ventilation) directly against safe hours — which is done in Part 3 after feature engineering.

8. Multivariate Analysis

A Pearson correlation heatmap was built at the patient level using mean vital values per patient, demographic features, and a ventilated binary flag. This provided an initial estimate of feature utility and potential multicollinearity.

8.1 Correlations with Ventilation Status

Feature	Correlation with Ventilated	Interpretation
mean_fio2	0.11	Strongest — higher O2 requirement linked to ventilation
mean_hr	0.04	Weak — slightly elevated HR in ventilated patients
age	0.03	Near-zero — age is not a strong linear predictor
admissionweight	0.00	No linear relationship
admissionheight	-0.01	No linear relationship
mean_sao2	0.00	Flat at patient-level averages

IMPORTANT NOTE

Low correlations at the patient level do NOT mean features are not useful. Pearson correlation measures only linear relationships on aggregated data. The real deterioration signal is non-linear and time-local — visible only in the 6-12h danger window, not in patient-level averages. Tree-based models (XGBoost, Random Forest) capture these patterns far better.

8.2 Feature Intercorrelations

- **Weight and Height:** Moderate positive correlation (0.25) — taller patients tend to be heavier. Expected.
- **Heart Rate and Respiration Rate:** Moderate positive correlation (0.28) — tachycardia and tachypnea often co-occur in sick patients. Both retained as they capture different physiological processes.
- **SpO2 and FiO2:** Negative correlation (-0.28) — clinically perfect. Patients who need more supplemental oxygen have lower SpO2. This is the main oxygenation signal.

- **No high multicollinearity:** No feature pair exceeded 0.8 correlation. No features need to be dropped on multicollinearity grounds.

9. Feature Engineering

Feature engineering transformed the raw, sparse event-based tables into a dense hourly grid suitable for supervised machine learning. Every engineering decision was driven by either EDA findings or clinical reasoning.

9.1 Building the Hourly Grid

Each patient's ICU stay was expanded into one row per hour, from hour 0 to the discharge hour. The variable t_min represents the start of each hour in minutes since ICU admission.

- Patients with unitdischargeoffset = 0 (12 patients) were excluded — a zero-duration stay cannot produce a valid timeline.
- The expansion produced a base grid of approximately 136,879 rows across 2,508 patients before label filtering.

9.2 Label Creation and Leakage Prevention

The binary target y_high was created by merging the earliest ventilation start time per patient into the grid:

- **y_high = 1** if t_min is in [vent_start_min - 720, vent_start_min - 360] (12 to 6 hours before ventilation)
- **Leakage removed:** Rows where t_min > vent_start_min - 360 (last 6 hours) and rows at or after vent_start_min were dropped entirely.
- **vent_start_min dropped:** The ventilation start time column was removed from the final dataset before modeling, as it is directly derived from the target variable.

9.3 Feature Alignment via merge_asof

All time-series features were aligned to the hourly grid using pandas merge_asof with direction='backward'. This function finds the most recent reading at or before each patient-hour — equivalent to carrying forward the last known clinical state.

This is preferred over regular merging because:

- Vitals are not recorded every hour. A regular merge would produce NaN for most rows.
- Forward-filling a merge_asof result matches what a clinician would observe: the most recent charted value.
- The backward direction ensures no future data ever enters the feature set — eliminating this source of leakage entirely.

CRITICAL SORTING NOTE

merge_asof requires both tables to be sorted by the merge key (t_min) only — not by patientunitstayid first. Sorting by patient ID first causes a ValueError. The by='patientunitstayid' parameter handles patient grouping separately from the sort order.

9.4 Features Aligned via merge_asof

Feature	Source Table	Signal	EDA Justification
heartrate_last	vitalPeriodic	Last known HR	99.6% complete; elevated in danger hours (90.7 vs 85.3 bpm)
sao2_last	vitalPeriodic	Last known SpO2	88% complete; lower in danger hours (95.8% vs 96.6%)
respiration_last	vitalPeriodic	Last known RR	85% complete; higher in danger hours (22.3 vs 20.0 br/min)
fio2_last	respiratoryCharting	Last known FiO2	96% vent coverage; highest correlation (0.11); 47.2% vs 41.1%
peep_last	respiratoryCharting	Last known PEEP	86% vent coverage; pre-intubation clinical escalation signal
on_oxygen_therapy	treatment	Was O2 therapy active	74% of ventilated patients; clinical escalation flag
on_noninvasive_vent	treatment	Was NIV active	23% of ventilated patients; BiPAP/CPAP before intubation

9.5 Trend Features

Single-timepoint readings are weak predictors. Deterioration is characterised by trends, not absolute values. A patient with heart rate 95 bpm is ambiguous; a patient whose heart rate has risen from 70 to 95 over three hours is clearly worsening.

Three trend statistics were computed using a 3-hour rolling window per patient:

Statistic	Variable Name Pattern	Clinical Meaning
Rolling mean (3h)	_mean_3h	Smoothed average — reduces noise from single-reading spikes
Rolling standard deviation (3h)	_std_3h	Variability — unstable vitals suggest loss of homeostasis
Lag difference (3h)	_slope_3h	Direction of change — is the patient getting better or worse?

Applied to heartrate_last, sao2_last, and respiration_last — producing 9 additional trend features. Part 3 EDA confirmed that respiration_mean_3h and sao2_slope_3h had the highest correlations with y_high among all features.

9.6 Patient Demographics

Static patient features were merged from the patient table. They do not change over time and are repeated for every row of the same patient.

10. EDA on the Engineered Dataset

After building the full hourly grid with all features, a second round of EDA validated that the features are genuinely predictive of y_high. This step — comparing danger hours directly against safe hours — is more informative than the raw-data bivariate analysis because it operates on the same time resolution as the model.

10.1 Feature Means: Safe vs Danger Hours

Feature	Safe Hours (y=0)	Danger Hours (y=1)	Direction	Clinical Interpretation
heartrate_last	85.3 bpm	90.7 bpm	↑	Tachycardia 6-12h before ventilation
sao2_last	96.6%	95.8%	↓	Early hypoxia — small but consistent drop
respiration_last	20.0 br/min	22.3 br/min	↑	Tachypnea: respiratory distress signal
fio2_last	41.1%	47.2%	↑	Increased supplemental oxygen requirement
peep_last	5.47 cmH2O	5.33 cmH2O	~	Similar — PEEP readings mostly after vent
heartrate_slope_3h	-0.14	-0.58	↓	HR falling slightly faster — unclear signal
sao2_slope_3h	-0.04	-0.21	↓	SpO2 declining faster in danger hours
on_oxygen_therapy	0.47	0.41	—	Present in both groups — context flag

VALIDATION

All physiologically meaningful features move in the clinically expected direction during danger hours. Higher FiO2 need, faster respiration, elevated heart rate, and falling SpO2 trends all precede ventilation. This confirms that the feature engineering captured genuine pre-ventilation clinical signal.

10.2 Feature Correlations with y_high

Pearson correlations between features and y_high were computed on the full engineered dataset. Top results:

Feature	Correlation	Note
respiration_mean_3h	+0.020	Trending RR strongest — confirms trend features add value
respiration_last	+0.018	Single-reading RR also predictive
fio2_last	+0.016	EDA prediction confirmed at row level
heartrate_mean_3h	+0.015	Trending HR more predictive than single reading
sao2_missing	+0.014	Missing SpO2 is informative signal
hr_missing	+0.014	Missing HR is informative signal
sao2_mean_3h	-0.013	Lower SpO2 trend in danger hours
sao2_last	-0.012	Slightly lower SpO2 in danger hours

The low absolute values (maximum 0.02) are expected and do not indicate poor features. With only 0.22% positive examples, Pearson correlation is heavily suppressed. Tree-based models such as XGBoost will identify these non-linear, interaction-based patterns far more effectively.

11. Key Findings & Feature Decisions

Feature	Source	Decision	Evidence
heartrate_last	vitalPeriodic	INCLUDE	99.6% complete; 5.4 bpm higher in danger hours
sao2_last	vitalPeriodic	INCLUDE	88% complete; lower in danger hours; SpO2 is primary O2 monitor
respiration_last	vitalPeriodic	INCLUDE	85% complete; 2.3 br/min higher in danger hours
fio2_last	respiratoryCharting	INCLUDE	96% vent coverage; highest raw correlation (0.11)
peep_last	respiratoryCharting	INCLUDE	86% vent coverage; key pre-intubation clinical signal
*_mean/std/slope_3h	engineered	INCLUDE	Trend features outperform single readings in EDA

Feature	Source	Decision	Evidence
age, weight, unittype, gender	patient	INCLUDE	Clinical context; potential interactions with vitals
on_oxygen_therapy	treatment	INCLUDE	74% of ventilated patients; escalation flag
on_noninvasive_vent	treatment	INCLUDE	23% coverage; BiPAP/CPAP as pre-intubation step
hr/sao2/fio2_missing	engineered	INCLUDE	Missingness itself is clinically informative
dischargeweight	patient	EXCLUDE	51% missing; post-ICU — not available at prediction time
temperature, cvp, etco2...	vitalPeriodic	EXCLUDE	87-99% missing; requires invasive monitoring
RR (patient) from resp_chart	respiratoryCharting	EXCLUDE	35% vent coverage; duplicated by vitalPeriodic
LPM O2, bronchodilator	multiple	EXCLUDE	< 36% and 19% vent coverage respectively
vent_start_min	engineered	EXCLUDE	Directly related to y_high — severe data leakage

12. Next Steps — Preprocessing & Modeling

The EDA phase produced ventilation_final.csv: 136,867 rows, 28 features, y_high label, 0.22% positive rate. The following steps are planned for Notebook 2.

12.1 Preprocessing Rules

- Patient-level train/test split (80/20) — rows from the same patient must not appear in both sets. Row-level splitting would allow the model to memorise patients rather than learn generalisable patterns.
- Fit all transformers (imputer, encoder, scaler) on the training set only. Apply fitted transformers to the test set. Never let test data statistics influence training.
- Impute remaining NaN values with median (numeric) and mode (categorical) from train.
- One-hot encode unittype and gender. Handle unknown categories with handle_unknown='ignore'.
- StandardScaler for continuous numeric features. Binary and one-hot columns are not scaled.

12.2 Class Imbalance

Two strategies will be tested:

- **class_weight='balanced':** Penalises misclassification of the positive class more heavily. No data modification. Simpler and preferred as a starting point.
- **SMOTE:** Creates synthetic positive examples by interpolating between real danger-hour rows. Applied only to the training set. Used if class_weight alone produces insufficient AUPRC.

12.3 Models

Model	Role	Imbalance Strategy	Key Parameters
Logistic Regression	Baseline	class_weight='balanced'	max_iter=1000
Random Forest	Ensemble	class_weight='balanced'	max_depth=10, min_samples_leaf=50
XGBoost	Expected Best	scale_pos_weight=~459	n_estimators=300, eval_metric='aucpr'

12.4 Evaluation

PRIMARY METRIC: AUPRC

Area Under the Precision-Recall Curve. The correct metric for extreme class imbalance. A random classifier achieves AUPRC = positive rate = 0.0022. Any meaningful model must substantially exceed this baseline. AUROC is reported as a secondary metric for comparison with published literature.

End of EDA Phase Report — Output: ventilation_final.csv — 136,867 rows — 28 features — y_high label